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Informatica® Cloud Application Integration
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Pinecone Connector Guide

Informatica Cloud Application Integration Pinecone Connector Guide
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Preface

Read the *Pinecone Connector Guide* to learn how to set up and use Pinecone connections. This guides assumes that you have an understanding of the Pinecone API concepts.

CHAPTER 1

Pinecone Connector overview

Pinecone Connector is a set of tools and APIs designed to integrate Pinecone's vector database capabilities with various data sources, platforms, and applications. Pinecone Connector streamlines the process of ingesting data into Pinecone, performing vector searches, and integrating search results into your application workflows.

You can use Pinecone Connector to achieve your general use cases, such as building search engines, developing recommendation systems, creating chatbots and virtual assistants, implementing personalization features, and performing data analysis to gain insights.

Pinecone Connector provides the following features:

- Supports the ingestion of data from various sources into Pinecone.
- Provides methods for querying Pinecone indexes that allow you to perform similarity searches and retrieve relevant results based on vector embeddings.
- Allows you to configure and fine tune search parameters, including similarity metrics and search algorithms.

Introduction to Pinecone

Pinecone is a cloud-native vector database service designed to simplify the development of AI-powered applications. It provides a platform for storing, searching, and managing high-dimensional vector embeddings that are crucial for various modern AI and machine learning tasks.

Pinecone provides the following key features:

- Offers a simple and intuitive API for integrating vector search into your applications.
- Manages high-dimensional vectors that are used to represent complex data such as text, images, or user behaviors in a form that machines can process.
- Provides efficient and scalable similarity search capabilities.
- Automatically scales to accommodate growing datasets and query volumes.
- Optimized for high performance with features, such as distributed computing and vector similarity search algorithms, to ensure fast query responses and support high-throughput use cases.

CHAPTER 2

Pinecone connections overview

Use the Pinecone REST APIs and HTTP requests to interact programmatically with a Pinecone account.

You can use the Pinecone REST APIs to perform the following tasks:

- Query, fetch, update, upsert, or delete vectors
- Get index statistics
- Get a list of vector IDs and indexes

After you create a Pinecone connection, you can validate, test, and save the connection.

You can then publish the Pinecone connection and click the **Metadata** tab to view the generated process objects for the connection.

Basic connection properties

The following table describes the basic properties that you can configure on the **Properties** tab of the connection creation page:

Property	Description
Name	Required. Unique name for the Pinecone connection that identifies it in the Process Designer. The name must start with an alphabet and can contain only alphabets, numbers, and hyphens (-).
Location	The location of the project or folder where you want to save the connection. Click **Browse** to select a location. If the **Explore** page is currently active and a project or folder is selected, the default location for the connection is the selected project or folder. Otherwise, the default location is the location of the most recently saved asset.
Description	Optional. Description of the connection.
Type	Required. The type of connection you want to use for the connector or service connector. Select **Pinecone**.
Runtime Environment	Required. The runtime environment for the connection. You can run the connection on the Cloud Server, a Secure Agent group, or a Secure Agent machine.
Connection Test	Not supported for Pinecone Connector.
OData-Enabled	Not supported for Pinecone Connector.

Pinecone connection properties

To make calls to the Pinecone API, you must provide a valid API key for the Pinecone project. Enter the API key in the **API_Key** property in the **Connection Properties** section. The Pinecone API uses the API key to authenticate Pinecone connection requests.

Create a Pinecone API key

Create a Pinecone API key to access the Pinecone API. You need an API key to make API calls to your Pinecone project.

All API calls to your Pinecone index authenticate with an API key for the project containing the target index. If you are using a Pinecone client, you can initialize a client object that allows you to provide your API key in one place and use it multiple times. If you are making HTTP requests, the HTTP request must include a header that specifies the API key.

To create a Pinecone API key, perform the following steps:

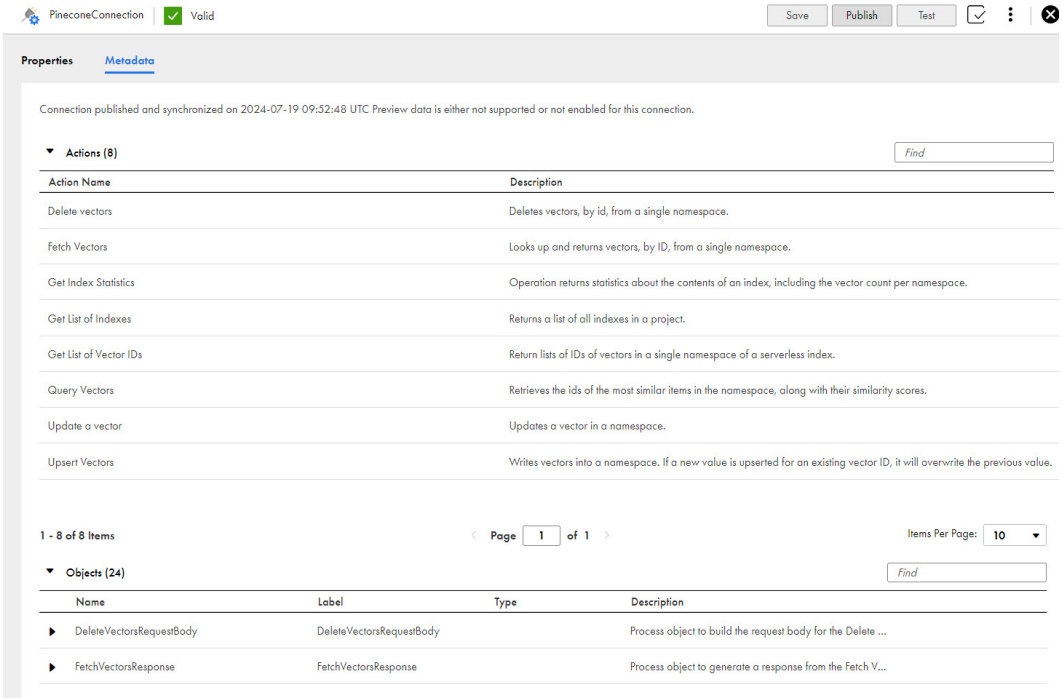
1. Open the Pinecone console.
2. Select your project.
3. Go to **API Keys**.
4. Copy the API key.

Pinecone connection metadata

After you create a Pinecone connection, you can validate and save the connection. You can then publish the Pinecone connection and click the **Metadata** tab to view the generated process objects for the connection.

When you publish a Pinecone connection in Application Integration, a list of **Actions** and **Objects** appears on the **Metadata** tab.

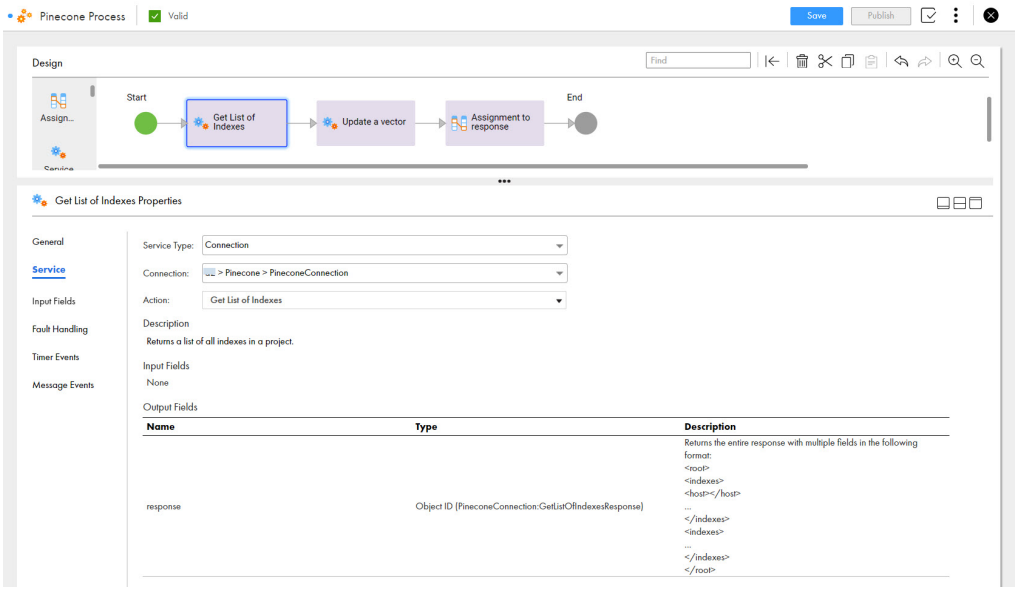
The following image shows the **Metadata** tab for a published Pinecone connection:



Using a Pinecone connection in a process

After you create a Pinecone connection, you can use it in a Service step in a process.

1. Add a Service step to the process.
2. Click the **Service** tab.
3. From the **Service Type** list, select **Connection**.
4. From the **Connection** list, browse and select the Pinecone connection that you created.
5. From the **Action** list, select the action that you want to perform. For example, select the **Get List of Indexes** action to return a list of all indexes in a project as shown in the following image:



In this case, Application Integration populates the **response** object under the output fields. Similar to the Get List of Indexes step, you can configure more service steps as required.

6. Configure further steps as needed.
7. Validate, save, and publish the process.

Rules and guidelines for using Pinecone connector

Consider the following rules and guidelines when using the Pinecone connector:

- To use the Fetch Vectors action, you must add array of IDs to the query parameters as shown in the following sample code:

```
<root>
<element>id-1</element>
<element>id-2</element>
<element>id-3</element>
<element>id-4</element>
</root>
```

- The Get Index Statistics, Query Vectors, and Delete actions might contain filter properties. You might create a requestBody with a filter field. Filter field uses query language with query selectors. To build a filter field correctly, you must replace the `$` with the `_`. For example, replace `$eq` with `_eq` as shown in the following sample code:

```
<root>
    ...
    <vector>-0.120900110</vector>
    <filter>
        <genre>
            <_eq>comedy</_eq>
        </genre>
    </filter>
    <topK>3</topK>
</root>
```

For more information about filter data with metadata, see Pinecone documentation.

- To use fields with data types, such as array, boolean, double, or integer, you must use annotation as shown in the following sample code:

```
<root xmlns:m="urn:informatica:ae:xquery:json2xml:meta-data">
    <field_array m:isArray="true">
        <property_1> ...</property_1>
    </field_array>
    <field_boolean m:type="xs:boolean">true</field_boolean>
    <field_double m:type="xs:double">10.3</field_double>
    <field_integer m:type="xs:int">5</field_integer>
</root>
```

- If you need additional types in the field, use the additional namespaces as shown in the following sample code:

```
<root xmlns:m="urn:informatica:ae:xquery:json2xml:meta-data">
    <vector>0.3</vector>
    <vector>0.3</vector>
    <vector>0.3</vector>
    <vector>0.3</vector>
    <vector>0.3</vector>
    <vector>0.3</vector>
    <vector>0.3</vector>
    <vector>0.3</vector>
    <filter>
    <_and>
        <genre>
            <_eq>drama</_eq>
        </genre>
    </_and>
    <_and>
        <year>
            <_gte m:type="xs:int">2020</_gte>
        </year>
    </_and>
</filter>
<topK>3</topK>
<includeValues>true</includeValues>
</root>
```